

Title: Environmental effects overtake selection to shape avian body size
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Abstract

Natural selection widely favours the largest within animal species. Why body sizes are declining in the world's temperate birds is therefore mechanistically confusing and ecologically worrying. We argue that understanding the causes of these declines requires knowledge of how complete, body mass probability distributions – not just averages – are moving. Using data from 159 North American bird species – spanning 26 families, 53° of latitude, and 25 years – we show that body mass distributions are not only downgrading, but also narrowing, and losing their upper tails (the heaviest individuals), with the largest changes occurring at southern range limits. Combining fitness models with a novel, distributional Price equation revealed that these changes are increasingly driven by phenotypic plasticity, not selection or genetic drift, and may yet be reversible.